

Preverbal *ge-* in Old and Middle English: Resultativity, Aktionsart and Viewpoint Aspect

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1 Background

1.1 Introduction

This paper aims to advance our understanding of the role of preverbal *ge-* in Old English (henceforth OE) and its disappearance in Middle English. This prefix is reminiscent of its cognates in Contemporary German and Dutch¹ in its distribution, but even a cursory examination of the details reveals the OE prefix (and indeed its cognates in the other Old Germanic languages) to be quite distinct. The proper characterization of that distribution, of the morphosyntactic and semantic role the prefix played in the language, and of its diachronic development, has proven to be extremely difficult. I propose to approach these questions by pursuing two strategies simultaneously. First, we can vastly improve the empirical basis for discussion by taking advantage of the parsed electronic corpora that have become available in recent decades. Second, our analysis can be profitably informed by taking into account developments in the theoretical treatment of verbal prefixes and of the array of related but distinct grammatical categories often confused under the label aspect, and of how these categories are marked in other ways in OE and ME.

I have thus carried out a large-scale corpus study using the *York-Toronto-Helsinki Parsed Corpus of Old English Prose* (Taylor et al., 2003, henceforth

¹These are also orthographically *ge-*, but have quite different pronunciations due to the effects of regular sound changes in the three languages. While (Standard) German has [gə], and Dutch has [χə], the Old English pronunciation was something like [je] or [jə]

YCOE)² and the *Penn-Helsinki Parsed Corpus of Middle English*, 2nd ed. (Kroch and Taylor, 1999, henceforth PPCME2).³ This has yielded rather large quantities of data — 217,326 clauses for OE and 163,599 clauses for ME, each of which has a main verb form that could in principle be prefixed with *ge-*. The fact that these corpora are parsed and annotated for syntactic structure means that we can code each of these examples for a whole series of morphosyntactic and lexical properties to see which correlate with the presence or absence of the prefix. This paper reports the results that can be gleaned from a series of automated searches run on these data. The patterns that emerge then serve as the empirical basis for an improved description of the facts about the distribution of *ge-* over the course of OE and ME.

Building on this, I develop a more precise theoretical hypothesis about the morphosyntax and semantics of *ge-* in OE than any found in previous work. This hypothesis not only explains the broad outlines of distribution of the prefix, but also covers some otherwise surprising points of detail involving particular lexical verbs, clarifies how it relates to the marking of Viewpoint Aspect and Aktionsart in other languages, makes sense of how it develops from OE to ME, and even provides insight into why it appears as a prefix rather than a suffix. I will propose specifically that *ge-* in OE was the default realization of a *res(ultative)* head in the sense of Ramchand (2008). In other words, rather than marking an aspectual category or a specific Aktionsart, it was one way to realize a particular piece of structure that contributes a component to the determination of Aktionsart — an extremely central component to be sure, but a quite specific one.

It is important to note at the outset that the results I present here are incomplete in a very particular way. As I will discuss in some detail in Section 2.1, the specific strategy pursued for the corpus work produces an extremely large amount of data, but at the cost of somewhat reduced accuracy and, above all, without the possibility of an examination of the individual examples by a human researcher. It is thus to be hoped that the current study can be complemented by work adopting a version of the opposite strategy. This would involve a close reading of a manageably sized, representative sample of relevant clauses from the corpus, examining in particular details of interpretation and other subtle properties that are not reflected in the corpus annotation and cannot be searched for electronically.

²Approximately 1.5 million words, <http://www-users.york.ac.uk/~lang22/YCOE/YcoeHome.htm>

³Approximately 1.2 million words, <http://www.ling.upenn.edu/hist-corpora/>

1.2 Germanic comparisons

The prefix *ge-* is the descendant of a common Germanic element **ga/gi*, the cognates of which are found in most of the other (Old) Germanic languages. As we will discuss in detail in Section 1.4 below, the traditional analysis of the original prefix in its preverbal use in Proto-Germanic is that it marked something like perfectivity or resultativity (Streitberg, 1891; Lloyd, 1979; Wedel, 1997; van Kemenade and Los, 2003, and many others), but our understanding is complicated by the notoriously difficult nature of aspectual categories and the distinct developments it has undergone in the daughter languages.⁴ I will consider the languages in a somewhat idiosyncratic order that is helpful for building up the discussion.

I begin with the North Germanic branch, because this can be dealt with most quickly. The reflex of **ga/gi* was lost prehistorically in these languages, just like the other unstressed verbal prefixes, so that by the Old Norse period (henceforth ON) it no longer appears as an identifiable prefix. More specifically, pre-tonic unstressed vowels underwent syncope, and most of the resulting consonant clusters were subsequently simplified by deleting the first consonant. This effectively eliminated the entire prefix in most instances. In cases where allowable initial clusters would have resulted, the consonant portion of the prefix was nonetheless usually eliminated analogically. It remained as a relic in a handful of cases, where the origin of the consonant was presumably obscured, and speakers would almost certainly have regarded it as part of the verb root rather than an identifiable prefix. Examples include ON *glíkr* ‘gleich’ cf. Gothic *galeiks*, and ON *gnógr* ‘sufficient’ cf. Gothic *ganōhs*. See Noreen (1923) for full discussion and additional examples.

In Gothic, *ga-* was a prominent part of the morphosyntax of the language. Its distribution shows strong similarities to what we find in OE and the Old West Germanic languages, though the relatively small corpus available to us and concerns about translation effects make it difficult to achieve a fully adequate understanding of its exact role. Nonetheless, there is an extensive (mostly older) literature attempting to do just that (see especially Streitberg, 1891; Lloyd, 1979; Eythórsson, 1995; Wedel, 1997, for

⁴The prefix could in fact attach to nouns, adjectives and even adverbs in the older languages, and the different uses clearly have a common origin. However, this lies quite far back in time, perhaps predating Proto-Germanic. Certainly by the time of the historically attested languages the connection between them is rather tenuous, so that it no longer makes sense to propose a unified syntax/semantics for them. I will thus focus here solely on the preverbal uses of the prefix and set the other ones aside. See Arista (2012) for some discussion of the relationships between the different uses of *ge-* in OE and Germanic.

data and discussion). The consensus is that what *ga-* was doing was somehow related to aspect, though the details are controversial and somewhat obscure, as we will see below.

In the Old West Germanic languages, the prefix was well preserved, again playing an active role in the verbal morphology. In OE in particular, *ge-* was still an obviously productive part of the verbal system, with a behavior similar to that in Gothic, which has eluded straightforward characterization. It is clear that it interacts with aspect, Aktionsart, and (lexical) semantics, but as is often the case when such intimately related factors are involved, it's far more difficult to say which of these factors defines its primary function — especially when we don't have access to native-speaker intuitions to help sort things out. Matters are not helped by the fact that *ge-* was lost as a productive element in the course of the Middle English (henceforth ME) period, as I will discuss in some detail in Section 4. Present Day English (henceforth PDE), like ON, has only traces of *ge-*, where its origins as a prefix have become obscure, and even that mostly in non-verbal uses (e.g. the *e-* in *enough* and the *a-* in *aware* and (possibly) *alike*). This means that we can get no clues from the contemporary language, where native speaker intuitions would have been available to test particular hypotheses.

Contemporary German and Dutch differ from PDE, having preserved the prefix as a productive element of their verbal morphology up to the present day. This means that we do have access to living native speakers here, so that we can achieve a very accurate description of the distribution of *ge-* in these languages. However, this doesn't do much to help us get at the nature of the prefix in OE or even Old High German (henceforth OHG), because German and Dutch have clearly undergone significant innovations in this area. While we find at first glance similarities between the three languages in the behavior of *ge-*, a quick comparison shows crucial differences in the details, meaning that the German and Dutch patterns would be a poor starting point for investigating OE.

My discussion here will be based on the German facts, but the characterization applies in its essentials to both languages.⁵ In Contemporary German and Dutch, the prefix has two distinct verbal functions. First, it appears productively on the participial form of the verb which is used (along with various auxiliaries) to form the periphrastic perfects and passives and can also appear attributively (henceforth the PPP, for perfect/passive participle). In German, this is restricted to verbs with stress on the first

⁵The distribution of *ge-* is not identical in the two languages (see fn. 6), but the differences (as far as I am aware) are not relevant to the points being made here.

syllable, but is otherwise completely regular and productive.⁶ If the verb is stressed on a non-initial syllable — either because it has a different unstressed prefix or a multisyllabic root with non-initial stress — then no *ge-* prefix is added. Compare (1a) to (1b):

- (1) a. **Initial stress:**
- | Inf. | | PPP | Gloss |
|----------|---|-----------|----------|
| 'zählen | ~ | gezählt | 'count' |
| 'trinken | ~ | getrunken | 'drink' |
| 'mailen | ~ | gemailt | 'e-mail' |
- b. **Non-initial stress:**
- | Inf. | | PPP | Gloss |
|------------|---|----------|--------|
| er'zählen | ~ | erzählt | 'tell' |
| spa'zieren | ~ | spaziert | 'walk' |

Second, it appears non-productively as a derivational prefix, as in the examples in Table 1. Note that there is no consistent semantic relationship between such *ge-* prefixed verbs and their non-prefixed counterparts, and in some cases it is obscure what connection there could be between them at all. This of course represents a fairly typical pattern with derivational

<i>brauchen</i>	'need, use'	<i>gebrauchen</i>	'use'
<i>fallen</i>	'fall'	<i>gefallen</i>	'please'
<i>hören</i>	'hear'	<i>gehören</i>	'belong to'
<i>denken</i>	'think'	<i>gedenken</i>	'commemorate'
<i>stehen</i>	'stand'	<i>gestehen</i>	'confess'

Table 1: German derivational *ge-*

morphology. When used productively, such morphology usually makes a consistent semantic contribution, but over time derived forms can become 'lexicalized', i.e. the connection to their original constituent parts can be weakened or lost, with the lexical item that was once productively derived henceforth following a distinct path of semantic development from its parts. Consider in this connection examples like English *transmission* (which, in addition to a compositional meaning 'event of transmitting' has a far more opaque meaning that describes a central system in motor vehicles). These have played an important role in the theoretical discussion

⁶The relevant condition is different in Dutch, where it is not stress-placement, but the presence or absence of a competing inseparable prefix that matters. So e.g. verbs like *telefoneren* 'telephone', with stress on the *-er-* suffix, have PPPs like *getelefoneerd* with *ge-*, whereas analogous German verbs like *telefonieren* have PPPs like *telefoniert* without *ge-*.

of derivational processes going back to at least [Chomsky \(1970\)](#); [Aronoff \(1976\)](#). This process of lexicalization seems to be especially favored when the derivational morphology involved ceases to be used productively in the language, as is the case in the German pattern at hand. While some of the other unstressed prefixes like *ver-*, *ent-* and *zer-* can still be used, at least occasionally, to create new verbs with reasonably predictable meanings, there is no such productive use of *ge-*.⁷

The interesting complication here is that, in addition to this derivational use, *ge-* has grammaticalized the inflectional use in the participles. Whatever account we may adopt of these developments in German and Dutch in particular, or of ‘lexicalization’ in general, it seems clear that we must distinguish the two uses of *ge-* just described. We have nothing to gain from attempting to unify, in our synchronic grammars of German and Dutch, the productive use in the formation of PPPs with the non-productive derivational use. Instead, a historical explanation of the homonymy we observe is appropriate. That is, we have here two prefixes that are pronounced the same because they represent divergent developments of a single prefix in the (pre-)histories of these languages, but no longer have any connection in the synchronic grammars.

A clue to the distribution of the original unified prefix comes from certain notable patterns in earlier historical stages of German. First, in OHG we still find semi-productive alternations in the presence of the prefix with certain verbs, showing a consistent semantic effect. E.g. we find *sizzen* ‘be sitting’ alongside *gisizzen* ‘sit down’. Second is the fact that, into the Middle High German period, the prefix is **not** used in the PPP of certain inherently perfective/resultative verbs like *quēman* ‘come’. See [Braune and Reiffenstein \(2004\)](#) on both points as well as [Lloyd \(1979\)](#); [Wedel \(1997\)](#) for additional data and discussion. We’ll see in Section 2.7 that this somewhat surprising fact is paralleled in Old English, and I’ll offer an account for it in Section 3.3. Indeed, the consensus of the literature is that, at least to first approximation, the role of *gi-* in OHG was at least similar to what we find in OE as well as Gothic (see [Lawson, 1965](#); [Lloyd, 1979](#); [Wedel, 1983, 1997](#), for data, discussion and older references).

Turning back to OE now, the main impression of similarity with Contemporary German and Dutch comes from the fact that, here as well, *ge-* quite typically shows up on PPPs, e.g. in periphrastic ‘perfect’ constructions:⁸

⁷So e.g. *sich ver-X* can be created to mean something like ‘to X in an erroneous fashion’, *ent-X* to mean ‘to de/un-X’, and *zer-X* to mean ‘to X to pieces’.

⁸Examples taken from the corpora in the Penn-York series are given with the source information from their ID tag in the corpus, which identifies the specific corpus, the source

- (2) ac hēo hæfde **ge**coren Crist hyre to brȳdguman
 but she had chosen Christ her to bridegroom
 ‘...but she had chosen Christ as her bridegroom.’
 (coaelive,+ALS_[Eugenia]:349.401)
- (3) forðan þe his gebedda **ge**faren wæs of līfe
 because his bedfellow gone was from life
 ‘...because his wife had passed away.’
 (coaelive,+ALS_[Maur]:131.1567)

However, unlike in German and Dutch, *ge-* is not an integral part of the PPP. A significant percentage of PPP examples lack the *ge-* prefix, even though the verb is morphologically compatible with it, as in (4).

- (4) sē gelēaffulla Oswold ... wæs Ø-cumen to Cynegylse
 the faithful Oswold ... was come to Cynegils
 ‘The faithful Oswold ... had come to Cynegils.’
 (coaelive,+ALS_[Oswald]:131.5455)

Furthermore, *ge-* is found with rather high frequency on other verb forms beyond the PPP, e.g. on the finite past form in (5):

- (5) Sē **ge**worhte ealle þing
 he created all things
 ‘He created all things.’
 (coaelive,+ALS_[Christmas]:66.51)

Crucially, *ge-* in examples like this does **not** look like the derivational *ge-* in German verbs like *gefallen* (though it may be similar to its ancestor). As we will see, it is far too frequent, too widespread across lexical verbs, and too regular in the interpretive restrictions on its appearance.

text and information on the page and sentence number from the published edition used. E.g. (2) comes from the YCOE, from Ælfric’s Lives of Saints, in particular the life of St. Eugenia, and is (part of) token 401 (tokens correspond roughly to sentences and are numbered sequentially through the sample for a particular text) found on page 349 of the print edition used in the creation of the corpus. I direct the reader to the documentation of the corpora for details on how to identify source details on the basis of the ID tags. I have converted the ASCII renderings used by the corpora back into the corresponding special characters, e.g. *+t* to *þ*, *+a* to *æ* etc. I have also taken the liberty of adding macrons to indicate vowel length where this is not indicated in the original texts.

1.3 On Viewpoint Aspect, Aktionsart and friends

In order to make sense of the OE and ME facts surrounding *ge-*, and of the various existing proposals that I will introduce shortly in Section 1.4, we need to settle on an understanding of and terminology for a complex of conceptual and grammatical categories referred to in the literature as aspect, Aktionsart and several other more idiosyncratic terms. I will attempt to take a position that is as neutral as possible, avoiding more contentious specific proposals, while also striving to be clear and explicit. I will distinguish two broad notions, which I will call Viewpoint Aspect and Aktionsart, choosing terms that make the distinction maximally clear.⁹ The capitalization is meant to signal when I mean these very specific understandings of the terms. I will reserve the bare term aspect for cases where I wish to be intentionally vague and general.¹⁰

Aktionsart is a property of a verbal predication, what we might regard as the complete VP or even vP. It is heavily constrained by the lexical verb, but is also determined by how this interacts with verbal complements of various categories. The Aktionsart of a VP characterizes the temporal and eventive structure of its description of an eventuality. Is the eventuality stative or eventive? Does it involve a non-trivial duration of time? Does it have a natural boundary or state of completion? Since Vendler (1957), there is a standard set of terms for four distinct Aktionsarten: States, Activities, Achievements and Accomplishments, to which have been more recently added Semelfactives. States involve duration and no eventivity, e.g. *Sally likes trucks*. Activities are durative events without a natural ending point, e.g. *Sally drives a truck*. Achievements are punctual transitions, i.e. they describe a change from one state to another e.g. *Sally started the truck*. Accomplishments describe something like an Activity, which however involves incremental change that culminates in a natural ending point, e.g. *Sally's truck climbed the mountain*. Semelfactives are punctual events that don't imply a change, and thus can easily be repeated iteratively, e.g. *Sally's truck belched smoke*. For present purposes, most interesting are Achievements and Accomplishments, because they

⁹Setting aside a host of even more idiosyncratic labels, what I'm calling Viewpoint Aspect is also often just called aspect, or outer aspect, or (confusingly, in older literature) Aktionsart. What I'm calling Aktionsart is also sometimes called inner or lexical aspect or action type.

¹⁰See e.g. Verkuyl (1972); Comrie (1976); Lloyd (1979); Klein (1994); Ramchand (2008) for relevant detailed discussion of the different ideas involved here, though the various sources may use very different terminology than I adopt. E.g. Comrie uses 'inherent meaning' and Lloyd uses 'actional types' and 'predicational types' to refer to roughly what I use the term Aktionsart for.

involve a natural ending point or state of completion. We say that they are **telic**, and that they include a notion of resultativity, specifically a **target state**, which is the ending point that holds as a result of the relevant eventualities reaching their natural point of completion.¹¹

Viewpoint Aspect is a property of the clause. It builds on Aktionsart and feeds into tense. The term I have chosen to refer to it is meant to indicate that it is based on the perspective from which the speaker chooses to view and describe the internal temporal and event structure of the eventuality. Because Viewpoint Aspect builds directly on Aktionsart, the two are intimately related, and it is not surprising that it has proven challenging to maintain a clear boundary between them. As we will see, particular Viewpoint Aspect categories can play out quite differently when combined with different Aktionsarten, and indeed certain combinations are essentially ruled out because of event-logical incompatibilities. There are several distinct Viewpoint Aspect categories in use in the world's various languages, which differ in details that we can't concern ourselves with here. I will restrict myself to a basic outline, distinguishing two main types. One type, traditionally called **imperfective**, involves the speaker viewing the eventuality from the inside, essentially setting up an imaginary present contained within the eventuality. The effect is the implication that the eventuality is ongoing or in progress. Being situated within an eventuality requires that the eventuality must have some duration, so this is incompatible with punctual Aktionsarten, or coerces them into things like habitual interpretations, where what is ongoing is not a single eventuality described by the basic predicate, but a period of time during which such an eventuality regularly occurs. Languages differ on this point in terms of what they allow, with some making further distinctions among broadly imperfective Viewpoint Aspects with respect to what it means for an eventuality to be ongoing.¹² The immersiveness of this kind of Viewpoint Aspect also means that the external outlines of the eventuality are not visible or accessible, so in these cases there is no implication of completion or resultativity, even if the basic predicate is inherently telic. The second type, traditionally called **perfective**, views the eventuality from the outside, as a complete delimited whole. In order for an eventuality to be viewable from the outside, it must have boundaries, hence perfectives imply completion in some sense. This does not necessarily imply resul-

¹¹See [Parsons \(1990\)](#); [Kratzer \(2000\)](#) for the difference between target states and resultant states.

¹²E.g. the English progressive is famously a type of imperfective that requires the ongoing eventuality to be eventive, hence is usually incompatible with pure statives: **The apple is being red*.

tativity or telicity, since an eventuality can come to a close for arbitrary reasons or be bounded by external factors, independent of whether it has a natural ending point or associated target state. Again, languages differ here, sometimes having multiple types of perfective constructions that place distinct additional restrictions. A central point of variation here is whether the eventuality is viewed from the outside as a monolithic and opaque whole, or with a kind of omniscience, where the speaker has access to both the external outlines and the internal structure of the eventuality. In the former case, there are no implications about the sub-eventive properties or the Aktionsart of the eventuality — it is simply reported that something happened. In the latter, it is possible to have additional implications relating to eventivity, duration, and above all telicity or resultativity of the eventuality. This latter point is crucial for some existing accounts of OE *ge-* and its cognates in the other Old Germanic languages.

1.4 Prior analyses

The wide range of rather tricky facts about the distribution of OE *ge-* and its cognates in the Old Germanic languages has led to an array of proposals about its meaning, function and grammatical status, which are usually quite abstract, often vague, and sometimes even completely empty. The most extreme view was that *ge-* was simply meaningless. According to Thomas Benson, *Vocabularium Anglo-Saxonicum* (1701) “*Ge-* apud Saxones semper fere superfluum” (‘Among the Saxons, *ge-* is almost always superfluous’). While later scholars have generally not adopted this view, they have repeatedly thought it worthy of mention as an indication of how difficult it is to pin the prefix down — as indeed I am doing here. Another approach, which was developed at length by Lindemann (1970), but has found little resonance, is that *ge-* expresses abstract direction. According to him, “the action expressed by any verb to which [*ge-*] is prefixed is directed toward some thing or in a direction forward and outward” [p. 37].

The most popular proposal, not just for OE, but also for its cognates in the other Old Germanic languages, is that *ge-/ga-/gi-* is a marker of (a kind of) perfective Viewpoint Aspect. The idea is associated in particular with Wilhelm Streitberg, who was inspired by comparisons with aspectual prefixes in the Slavic languages. He argued specifically that the Old Germanic languages had a parallel system, verbs with *ge-/ga-/gi-* or any of a series of further prefixes are perfective, and unprefixed verbs are imperfective (Streitberg, 1891, etc.). This view soon fell into disrepute, as it became clear that the distribution of the prefixes in Gothic and Germanic in

general was rather different in detail from the patterns observed in Slavic languages. Broadly similar approaches have been rehabilitated more recently, however, in work (e.g. by [Lloyd, 1979](#); [Eythórsson, 1995](#); [Wedel, 1997](#)) that attempts to take into account the differences in the aspectual systems of the two language families. The guiding idea here is that deviation from what is found in Slavic does not imply that what is involved is not Viewpoint Aspect. Rather, the details of the categories and contrasts in the various languages happen to be different, but they nonetheless involve distinctions at the same level, namely Viewpoint Aspect more or less as described above.

[Lloyd \(1979\)](#) in particular discusses the differences between Slavic and early Germanic aspect in great detail and argues in the end that a single system of primitives underlies both systems.¹³ They differ, however, in exactly which distinctions in those primitives they mark, and how they interact with other relevant factors of verbal predication like Aktionsart.¹⁴ First, the general set of primitives includes three distinct Viewpoint Aspect categories. Lloyd makes very clear that these differ most fundamentally in terms of the (imagined) perspective from which the individual he refers to as the observer/reporter describes a particular eventuality, and how this interacts with what they can report about the internal details of that

¹³It should be noted at the outset that [Lloyd \(1979\)](#) is quite idiosyncratic. The idiosyncrasy involves not only the use of terminology, but the entire landscape of analytic categories, especially when viewed from a present-day perspective based on how the field has developed in the interim. This is partly because he was drawing on earlier work on aspect in Indo-Europeanist and Germanicist traditions that hasn't been particularly influential in the subsequent development of the field (e.g. the term 'complexive' comes from [Hermann, 1927](#), and has had some currency in work on historical Indo-European languages, but is rarely encountered in synchronic theoretical or typological work on aspect). But it is also a consequence of the scope of the work. Rather than applying and modifying an existing theory of aspect to the Gothic material, Lloyd develops a whole new framework for conceiving of aspectual categories and contrasts from first principles, and then tests it out in comprehensive detail on the surviving Gothic material. His approach is heavily informed by and adopts a number of crucial ideas and distinctions from prior work on aspect, but the foundations are novel, based on essentially philosophical reflections on the nature of time, the properties of events in the real world, linguistic predication, and how predication can be applied to describe events. The result is quite impressive and full of original insights, but for better or worse it has had very little influence outside the discussion of aspect specifically in the Old Germanic languages. It is thus no wonder that it ends up being quite difficult to compare with other work.

¹⁴Lloyd avoids using the term Aktionsart, as it was hopelessly confused with Viewpoint Aspect, especially in much of the earlier work that he was building on. For what I am calling Aktionsart he uses the terms 'actional types' and 'predicational types', though I must confess that it remains unclear to me how these two are to be distinguished from each other.

eventuality. In the **constative** aspect, the observer/reporter views a non-present event from their vantage point in the actual present. As a result, they can only describe it as an essentially monolithic whole, “a delimited static fact of memory” [p. 73], without any consideration of what sort of internal structure it may or may not have. In the **presentive** aspect, the observer/reporter imagines themselves to be in a non-present time and reports on an eventuality essentially from the inside. Some internal structure is thus accessible, but the eventuality can only be presented as in progress or incomplete in some sense. The action cannot be described as complete, because this would require viewing it from the outside. Finally, the **complexive** aspect is when the observer/reporter describes an eventuality from a perspective that is outside of time, from which she can view all phases of a complete action. The term complexive is justified here because this allows presenting an eventuality not only as having finished, which the constative also does, but as having finished upon following some process to a natural ending point defined by the nature of the eventuality described.

Second, the Slavic and Old Germanic languages only make two-way distinctions in their formal means of expressing Viewpoint Aspect, and the two families map their two formal categories onto the three primitive aspectual categories in somewhat different ways. In Slavic, the aspectual system has two fully fledged categories of perfective and imperfective, and every verb form can be categorized as either the one or the other. The perfective is really just the complexive as described above, but Lloyd reserves the term perfective specifically for talking about the complexive as it is used in Slavic. The imperfective covers the presentive and the constative. In Gothic, *ga-* marks the complexive, which is thus essentially parallel to the Slavic perfective.¹⁵ However, the lack of *ga-* is aspectually neutral — it can be used for either the constative or the presentive, but it isn’t explicitly marked for any aspectual category. This means that Gothic doesn’t have a fully worked out Viewpoint Aspect system, so much as a way for one specific Viewpoint Aspect category to be marked. In Slavic, on the other hand, there is a fully developed system with two categories, where every verb form has a value for Viewpoint Aspect, either perfective or imperfective. There is no unmarked category, which plays out in the fact that, while the perfective is morphologically marked with many verbs — usually by means of a prefix — there are also imperfectives that

¹⁵An important development is that, while [Streitberg \(1891\)](#) proposed that all verbal prefixes marked perfectivity in Gothic, [Lloyd \(1979\)](#) restricts the complexive to *ga-*. The other prefixes mark certain actional types, i.e. a subset of what we would consider Aktionsart, which often leads to effects that are difficult to distinguish from perfective/complexive aspect, but is nonetheless separate.

are morphologically marked — typically with a suffix.

Third, building on the second point, there are important cross-linguistic differences in how the complexive aspect maps onto Aktionsart. In particular, what Lloyd calls ‘punctuals’, which essentially correspond to Achievements in the more familiar classification of Vendler (1957), are treated as ‘non-aspectual’ in Gothic, i.e. they don’t participate in aspectual contrasts, because they are incompatible with the complexive, and the complexive is the only Viewpoint Aspect that can be marked in the language. They thus never appear with *ge-* (Lloyd, 1979, p. 86f., 124). In Slavic, on the other hand, such verbs have acquired alternants that are explicitly marked as imperfective, with the result that the unprefixed forms have come to function as perfectives/complexives as a logical consequence of the duality in the system. In effect, whole classes of predications that can only be presented in constative aspect in Gothic, thus lacking the *ge-* prefix, are treated as perfectives/complexives in Slavic. This leads to a fundamental difference in how the complexive is used in Gothic vs. Slavic, and it is in part in response to this that Lloyd uses the more traditional term perfective to refer to the complexive in Slavic.¹⁶

The main alternative kind of approach, which I will argue is the most promising, is related to these proposals in terms of Viewpoint Aspect, but operates at a slightly different level. This is that *ge-* expresses resultativity in some specific way. While it is often true that perfectives are resultative, the converse is often not the case. That is, an inherently resultative Achievement predicate (like *break the window*) can quite easily be used imperfectively, e.g. with an additional progressive component (like *I was just breaking the window when the police arrived*). A role for resultativity in the prefix *ge-* has been proposed e.g. by van Kemenade and Los (2003); Los et al. (2012) for various stages of Dutch and English and Germanic in general,¹⁷ and this also connects to analyses of verbal particles in some of

¹⁶It is not entirely clear at what level we should conceive of this difference. One option is to say that the fundamental Viewpoint Aspect categories of Gothic and Slavic are somewhat different, thus complexive and perfective refer to distinct categories at every level. The other, which seems to be more in line with Lloyd’s discussion at various points in the book, is that at a fundamental level, there is a single universal definition of complexive aspect, which applies to Slavic and Gothic alike. The difference is that, in certain contexts where an eventuality could in principle be reported from either a constative or a complexive perspective, Gothic systematically chooses the constative, while Slavic systematically chooses the complexive, due to independent differences in the way that deep aspectual categories are mapped onto language-specific grammatical categories.

¹⁷Though note that their notion of resultative is broader than the one being assumed here. In particular, they regard predicates that involve a non-delimited change of state along a path as a type of resultative, so for them resultativity is e.g. orthogonal to telicity

the modern Germanic languages (see e.g. [Ramchand and Svenonius, 2002](#); [McIntyre, 2003](#)), where parallels are quite apparent. Special mention is due to [Karp \(1989\)](#), a rarely cited dissertation which concludes that, at least in *Beowulf*, “the *ge-* preverb consistently imparts a resultative meaning to a verb, “resultative” indicating the production of a result, the attainment of a goal or the change into a new condition, brought about through the action described in the verb stem” [p. 2]. This work is important for present purposes for two reasons. First, it is built around an extremely thorough and insightful investigation that covers every verb in the epic poem and thus has an extremely secure and deep empirical base that can nicely complement the broad but shallow base provided by the corpus study to be presented here. Second, Karp’s conclusion as to the function of OE *ge-* comes perhaps the closest of any previous work to what I will propose here. I follow her without qualification in the idea that *ge-* contributes resultative meaning, but will build on this by implementing the proposal in a more explicit theoretical framework, allowing for a more constrained understanding of resultativity that makes more specific predictions and can explain or at least accommodate a number of the wrinkles in the distribution of *ge-* that a simple resultativity story cannot.

1.5 Roadmap

In section 2 I will present in detail the findings of my corpus study on the distribution of our prefix in OE and in particular how it interacts with other identifiable properties of the verb and the clause containing it that might be expected to be relevant. Then in section 3 I will propose a specific version of the resultative analysis of *ge-*, taking it to be the default spellout of [Ramchand \(2008\)](#)’s *res* head. I will show how this can account not only for the broad facts of its distribution, but also for the small but consistent details that don’t fit with an analysis in terms of perfective Viewpoint Aspect or a naïve idea of resultativity. I turn in section 4 to developments in the ME period, when the prefix began to disappear from the language. While I will not be able to propose a clear answer to why *ge-* was lost, I will show that my analysis of its structural status in OE can provide some insight into how this loss interacted with other contemporary changes, in particular in the periphrastic perfect system. Section 5 concludes and sets the stage for future work.

([Los et al., 2012](#)).

2 The distribution of preverbal *ge* in OE

In this section I will present the main OE corpus study which will serve as the basis for the analysis developed in the rest of the paper. I will begin with a discussion of the particular methodology I adopted, and then work through a detailed presentation of the data, considering in turn how the distribution of *ge*- interacts with a series of factors annotated in the corpus.

2.1 Methodology

I am reporting here the results of a study of the complete YCOE [Taylor et al. \(2003\)](#), a corpus containing approximately 1.5 million words of OE prose, tagged for part of speech and other grammatical features and parsed for syntactic structure.¹⁸ The searches were carried out with the CorpusSearch program¹⁹, which was designed to search corpora in the Penn-Treebank format on the basis of lexical forms, POS tags and structural notions like dominance and precedence. This makes it possible to execute quite sophisticated searches efficiently.

In searching the corpus, I proceeded as follows. First, I ran searches to identify the main verb of each clause, since this is where preverbal *ge*- is primarily expected to appear.²⁰ Second, I classified each such main verb according to whether it was prefixed with *ge*-, with some other prefix which might be expected to be in competition, e.g. *be*- or *for*-, or not prefixed at all.²¹ Then I ran a series of searches to code each clause for properties that might be expected to have an effect on the distribution of

¹⁸In Section 4 I will report on a parallel study of the PPCME2 corpus for ME, using the same methodology described here.

¹⁹Version 2.003.00, <http://corpussearch.sourceforge.net>

²⁰The main verb is defined simply as the hierarchically lowest verb in the clause. The various (pre-)auxiliary verbs of the language never have *ge*- when used as auxiliaries (and essentially never do even when used as main verbs), so it is quite safe to exclude them. I have also elected, for the time being, to set aside participial verb forms used as attributive adjectives. This latter move is not necessarily innocuous, but was motivated by the need to keep the volume of data somewhat manageable.

²¹The logic here is fairly simple — with some very few exceptions, a single verb form will not bear two prefixes simultaneously, so *ge*- will be simply ruled out on a form prefixed with *be*- etc. This means that when we're trying to figure out the conditions on the appearance of *ge*-, a form prefixed with *be*- doesn't tell us the same thing as a form with no prefix. For the time being I have chosen to set aside the examples with other prefixes, and to focus on differences between forms with *ge*- and forms with no prefix, though I did consult the data on other prefixes in order to better understand the behavior of specific lexical verbs like *sendan* 'send', as we will see in Section 2.7.

ge-, either because they have been reported to do so in the previous literature, or because I could imagine a plausible connection to other factors that had been discussed. Finally, for each such property I checked to what extent it did in fact correlate with the choice between *ge-* and no prefix. As is standard practice, each round of CorpusSearch queries was interspersed with error checking and correction, both in the form of sanity checks with automated tools and examination by hand of manageable samples of individual examples. As we will see in the following subsections, there is a great deal of variation in whether and how much these factors actually correlate with the distribution of *ge-*, which provides clues about what the function of the prefix actually was.

As noted above, in doing this I was pursuing a particular strategy that took advantage of the nature of the phenomenon to yield an uncommonly large amount of data for the syntax of a historical language. This has obvious advantages, as we will see below, but it also involves a trade-off, as the sheer volume of data precludes certain types of analysis. Typically, in the corpora used here, examinations of specific grammatical phenomenon with a limited distribution might only turn up a few hundred or at most a couple thousand examples. These examples can be identified, categorized and coded by automatic searches in a first step, but once this is done, the data of interest is of a compact enough size that further investigative steps can be done ‘by hand’. That is, in a second step, the selected examples can feasibly be checked individually by a human researcher. This is key for confirming that the automated searches worked as intended, identifying occasional errors in the underlying annotation in the corpus, and the classification and coding yielded is correct in every case. More than that, it makes it possible to identify and tag more subtle properties in the examples, including interpretive ones, which are not reflected in the corpus annotation, and thus could not be searched for electronically. Such work is time consuming, but with numbers of examples in the hundreds or low thousands, it is feasible. In the present case, things look quite different. Every non-reduced clause in the corpora constitutes a relevant example, because every such clause has a main verb which either does or not have *ge-* prefixed to it. Just for the OE period this yields 217,326 examples. On the one hand, this lends a great deal of power to the analysis, meaning that even quite subtle and infrequent effects can be detected with a fair amount of accuracy. On the other hand, it makes it impractical to have human researchers check all of the automated coding, let alone do analysis of interpretive properties.

For the work presented in the current paper I thus chose to focus on what can be done with automated searching and coding of the full cor-

pora, with no attempt to systematically examine the individual examples by hand.²² This can be complemented by additional work that identifies a manageable subset of the data and subjects it to careful analysis by a human researcher for interpretive and other properties not reflected in the annotation. There is a sense in which Karp (1989) can be understood as work in this vein. It is restricted to *Beowulf*, setting aside all other OE texts, but examines every occurrence of a verb in the poem, in context, in great interpretive detail. I will indeed make use of it in this way to some extent in Section 3.2. There are advantages, however, to having the close reading of a subset of the data follow as a second stage after a first stage of automated analysis as reported on here. In this way, the first stage can inform and direct the second stage, suggesting specific interpretive and other properties that researchers should check the individual examples for in order to test specific predictions made by proposals emerging from the first stage.

2.2 The broad patterns

The first and simplest result of my searches of the corpus is that *ge-* is extremely common. Out of 166,544 clauses examined,²³ 42,366 (25.4%) had *ge-* on their main verb. Even setting aside PPPs, 30,862 of 153,622 main verbs (20.1%) had *ge-*. This is our first and perhaps clearest indication that *ge-* really did play an active and central role in the OE verbal system, and that it was quite unlike its Contemporary German and Dutch cognates. Its productive use is quite simply not restricted to the participial forms showing up in periphrastic perfects and passives, as it is in those languages.

The second broad result is that I have found some basic confirmation of previous claims: *ge-* tends to be favored in environments suggestive of perfectivity, telicity and resultativity, and disfavored elsewhere. At this rough level, it is difficult to distinguish among the various prior proposals, as there is a tendential relationship between perfectivity and resultativity.

²²Note that this does *not* mean there was no hand-checking of examples. As noted above, after each search or coding run with CorpusSearch, a sample of the output examples were examined to ensure that the automated query had worked as intended, with queries adjusted and rerun in case any errors were found. If done carefully, this can detect and correct the vast majority of systematic issues, yielding a good level of accuracy, though it will of course miss problems specific to individual examples outside of the samples examined by hand. More important, it does not allow for systematic examination of properties not reflected in the corpus annotation and thus not covered by the automated queries.

²³As noted above, this excludes all clauses whose main verb has a prefix other than *ge-*.

It is only when we consider certain details that a particular kind of resultative analysis begins to stand out as the most appropriate. Let us consider, then, the data on the different factors that are — or might have been expected to be — relevant to the distribution of *ge-*.

2.3 The form of the main verb

Given the fact that *ge-* is a crucial component of the productive formation of PPPs in Contemporary German and Dutch, we have reason to suspect that the specific form of the main verb will have some effect on the frequency of *ge-* in OE. We will see that this is indeed the case, and that here as well, *ge-* is extremely common on PPPs. However, a very important recognition is that, unlike in German and Dutch, *ge-* shows up in significant numbers on *all* morphosyntactically defined forms of the main verb. Since this is unfamiliar from the modern languages, it will be instructive to have examples demonstrating its appearance in each of them here.

First, we have the present participle, which corresponds semantically and syntactically most closely to the PDE form in *-ing*, but was formed in OE with the suffix *-nde* and most commonly combines with a form of auxiliary *be*.²⁴

- (6) & swā wæs **geendiende** þis wilwendlice liif
 and so was ending this temporal life
 ‘and thus [he] was ending this temporal life’
 (cobede, Bede_4:9.286.1.2881)

Then we have the *to*-infinitive, i.e. an infinitival form of the verb (in OE typically in the dative, ending in *-enne*), preceded by the preposition *tō*. As in PDE, these can appear as the sole verb of a non-finite clause or in a periphrastic construction with auxiliaries *have* or *be*.

- (7) and næfð nāne mihte [menn **tō gehælenne**]
 and not-has no power [men to heal]
 ‘and has no power to heal men’ (coaelhom, +AHom_4:86.569)

Next come finite verb forms, i.e. those marked for tense and agreement.

²⁴It should be kept in mind throughout this discussion that I am only considering occurrences of each of these forms when used as the main verb of a clause. Hence the relevant examples with the present participle are ones where it appears below an auxiliary, not e.g. cases where it appears in attributive position modifying a noun. The same goes *mutatis mutandis* for the other forms.

- (8) Sē **geworhte** ealle þing
 he created all things
 ‘He created all things.’ (coaelive,+ALS_[Christmas]:66.51)

Then we have bare infinitives, which typically show up in combination with the pre-modals and other auxiliaries, but are also found occasionally without auxiliary in certain non-finite clause types (see [Los, 2005](#), for detailed discussion of the different types of infinitive and their distribution in OE and ME).

- (9) þæt menn hit **gehȳran** mihton;
 that men it hear may
 ‘so that men may hear it’ (coaelhom,+AHom_1:451.233)

We also have examples of imperative forms of the verb, which show up in much the same contexts as in PDE.

- (10) and þonne þū eft cymst, **geoffra** þīne lāc.
 and when you again come, offer your sacrifice
 ‘and when you come back, make your offering’
 (coaelhom,+AHom_16:19.2269)

And finally we have the PPP, which occurs primarily in periphrasis with auxiliary *have* or *be* in the perfect or passive.

- (11) ac hēo hæfde **gecoren** Crist hyre tō brȳdguman
 but she had chosen Christ her to bridegroom
 ‘...but she had chosen Christ as her bridegroom.’
 (coaelive,+ALS_[Eugenia]:349.401)

All the same, even though *ge-* can show up on any form of the verb, its distribution across them is far from even. Rather, there are marked differences in its frequency on the various forms, as shown in Table 2, ordered from those occurring the least to those occurring the most frequently with *ge-*. What we see is that the distribution of *ge-* is skewed in the same direction as it is in German and Dutch, but not nearly so far. The prefix is extremely frequent with PPPs, but it is nowhere near categorical.²⁵

²⁵Note also that the situation here is quite different from the cases of PPPs without *ge-* in Dutch and German discussed above. In those languages, specific verbs are incompatible with *ge-* because they already bear another prefix (Dutch) or because they don’t have initial stress (German). The OE cases we are considering here are all ones without another prefix, and without any phonological incompatibilities with *ge-*. Indeed, we typically find PPPs of the same verb appearing *with ge-* in other contexts. We will see examples below.

form	ge-	no	% ge-
Pres. Ptc.	97	1503	6.1
to Infin.	430	2177	16.5
Finite	23707	102604	18.8
Bare Infin.	4325	11192	27.9
Imperative	2273	5468	29.4
PPP	11504	1418	89.0

Table 2: Form of the main verb

Similarly, it is extremely infrequent with present participles, but far from categorically absent. This is in line with what we might expect if *ge-* has to do with perfectivity or resultativity — perfects and passives, where the PPP mostly shows up, tend to be both perfective and resultative, while present participles tend to be used for the description of ongoing states or activities, which are generally imperfective and typically not resultative.

2.4 Tense, mood and negation

Given that the morphosyntactic form of the main verb is relevant for the distribution of *ge-*, we might expect this to carry over to more fine-grained distinctions like tense and mood. Indeed, if *ge-* has something to do with perfectivity or resultativity, it is plausible to think that tense in particular will make a difference, given well-known interactions between tense and aspect. E.g. in PDE, the progressive appears happily in both present and past, but the unmarked aspectual form is heavily restricted in the present, so that with eventive predicates a non-episodic (typically habitual) reading is forced.²⁶ This makes it quite difficult to use the construction with certain VPs.

- (12) a. I was eating the dosa.
b. I am eating the dosa.
- (13) a. I ate the dosa.
b. # I eat the dosa.

Example (13b) is odd out of the blue, because a habitual reading with a definite object requires a special context, e.g. as the answer to a question like ‘What do you usually have at this restaurant?’ On the other hand,

²⁶Lloyd (1979) has some nice discussions of such interactions, which motivate him to focus on the non-present tense for his definitions and discussion of Viewpoint Aspect categories. See in particular the aptly titled Chapter VI “Non-Present Actions and Aspect”.

the unmarked aspect is perfectly felicitous with an episodic reading in the past, as in (13a). Thus if OE *ge-* is involved somehow with Viewpoint Aspect, we might also expect it to interact with tense marking.

Looking at the numbers on the frequency of *ge-* according to the tense of the clause, we find that past tense has a small but perhaps genuine²⁷ favoring effect, as seen in Table 3.²⁸ This small effect can perhaps be un-

Tense	<i>ge-</i>	no	% <i>ge-</i>
Pres	15227	53528	22.1
Past	22888	59649	27.7

Table 3: Tense of the finite verb

derstood if *ge-* has something to do with the reaching of some result state. Whether or not a particular eventuality goes to completion is perhaps easier to judge and also more relevant when it lies in the past than in the present or future. Note relatedly that in many languages a perfective-imperfective distinction is restricted to or at least predominantly expressed in past forms, not present ones. However, given the relatively small effect size here, I will not make too much of this difference, relying more on clearer effects to motivate my proposals.

Mood seems less relevant. OE finite verbs distinguish subjunctives from indicatives, though a large number of forms are actually ambiguous between the two. Clear subjunctive forms have a somewhat higher frequency of *ge-* than clear indicatives, but ambiguous forms show the highest frequency, as can be seen in Table 4. There is reason to not take the differences here too seriously. That between indicative and subjunctive

²⁷I say “perhaps genuine”, because it appears to be statistically significant ($\chi^2 = 631.019$, $p < .0001$). However, the numbers of examples we are dealing with here and throughout this paper are so high that even extremely small differences will come out as (highly) statistically significant. This makes the results quite difficult to interpret in terms of consequences for our grammatical analysis. Sorting out how to interpret low effect sizes with high statistical significance will require a far more sophisticated statistical analysis, which can be better attempted at a later stage of research, once we have solid, testable hypotheses in place. Thus in what follows I will generally not report anything beyond basic descriptive statistics, except in the rare case where no significant effect could be detected even with such large volumes of data.

²⁸Note that what is at stake here is the tense of the clause, not necessarily of the verb form which we are considering with respect to whether it bears *ge-*. In the various periphrastic constructions, the tense of the clause will be marked on the (highest) auxiliary, whereas it is the main verb that we are examining for the presence of *ge-*. Of course, only finite clauses will have tense marked at all. The numbers here leave out the various untensed infinitival and participial clauses, thus do not add up to the same total as in some of the other tables, where all clauses are reflected.

	ge-	no	% ge-
Indicative	23031	75463	23.4
Subjunctive	5227	14720	26.2
Ambiguous	9857	22840	30.1

Table 4: Mood of the finite verb

especially is quite small in absolute terms, and the fact that ambiguous forms don't end up in between the two clear categories in terms of their prefixing behavior suggests that something else is going on here.²⁹

Another possibility I investigated is whether negation has an effect on the appearance of *ge-*. While the connection may seem obscure, [Postma \(2002\)](#) has shown that the cognate prefix *ghe-* in Middle Dutch actually had a preverbal use as a negative polarity item, so we might imagine that OE would exhibit something similar. Table 5 shows, however, that it does not. The difference we see here is tiny, and it goes in the opposite direction

	ge-	no	% ge-
Negative	3104	9812	24.0
Non-negative	39232	114550	25.5

Table 5: Polarity of the clause

of what Postma observed for Middle Dutch. I.e. rather than behaving as an NPI, *ge-* is *less* frequent in negated clauses than in non-negated ones.

2.5 Prepositional and adverbial elements

A standard class of diagnostic of aspect-related distinctions, in particular those at the Aktionsart level, is the licitness of certain PPs and adverbials. For example, we can classify predicates according to whether they can felicitously combine with PPs like *in an hour* or *for an hour* (roughly, telic vs. atelic predicates). Given the size of the corpus being examined here and

²⁹I.e. if there is a real difference between indicatives and subjunctives in their propensity to combine with *ge-* then, assuming that the ambiguous category contains a mixture of forms intended as indicatives and forms intended as subjunctives, it should show a behavior somewhere in between the two categories. What may be going on here actually is that there is an interaction with tense marking. Perhaps the largest group of forms where indicative and subjunctive are systematically syncretic in OE are the past 3sg forms of weak verbs. If syncretic forms tend to be past, then they will tend to take *ge-* at a higher rate for this reason, completely independent of any direct interactions between the prefix and mood.

the interpretive complexities involved, it was not feasible at this stage to divide things up according to specific PPs and adverbials. It is, however, relatively easy to search for whether a particular clause contains a clause- or VP-level PP or adverbial of any kind.³⁰ Consider then the frequencies for *ge-* under these conditions reported in Table 6. The prefix *ge-* is more

	ge-	no	% ge-
PP	20630	50821	28.9
No PP	21706	73541	22.8
Adverb	15456	43121	26.4
No adverb	26880	81241	24.9

Table 6: PPs and adverbs in the clause

frequent with both PPs and adverbials than without, though the difference is small. With adverbs it is just 1.5%, whereas with PPs there is a 6.2% increase, over four times the effect. The effect with adverbs is certainly small enough that, though again statistically significant, it may not be particularly meaningful (see the discussion in fn. 27 above). That with PPs may reflect something more real, but this cannot be determined until a detailed examination of individual examples in a sample of the corpus is carried out to see what kinds of PPs are associated with *ge-* and the lack thereof.³¹

³⁰The annotation of the YCOE and the PPCME2 does not indicate a VP or any other level between the clause and individual constituents likes NPs, PPs and APs, essentially because it would be impossible to reliably identify such levels without making a number of controversial analytical choices (and because there was, at least in the past, some controversy surrounding whether these languages were configurational, i.e. whether the notion of VP even played a significant role in their grammars). This means that there is nothing like a distinction in the annotation among different PPs and adverbials modifying different levels of the clausal spine. PPs and adverbials contained within APs and NPs, on the other hand, are clearly distinguished from those at the clause- and VP-levels, and these are excluded here.

³¹One possibility is that the frequency of *ge-* is increased by the presence of complement PPs which affect the Aktionsart of the main predicate, e.g. by introducing a telos, as in *Sandra swam* vs. *Sandra swam to the shore*. It is well known that such PPs can have a significant effect on auxiliary selection in Dutch and Italian, where telicity is a central factor (Sorace, 2000; van Hout, 2004; McFadden, 2007). The idea would be that such a complement role is more frequently played by PPs than by adverbials. Since complement vs. adjunct status of these elements is not consistently annotated in the corpus, this cannot be searched for automatically, but will have to be determined by examination of individual examples.

2.6 Date

Given the fact that *ge-* disappears in the course of the ME period, it is worth looking into whether it was already in retreat in OE. In other words, we want to see whether the frequency of *ge-* correlates with the date of a particular text. For the OE period it is exceedingly rare that we know exactly when a particular text was composed, or even when the surviving manuscripts (which are usually later, often considerably so) were written. The best we can do with a reasonable degree of certainty is typically a range of a few decades. Furthermore, as the collection of available texts is quite limited, if we tried to assign too narrow a date range to each, we would end up with unacceptably small amounts of data for any particular range. To deal with these issues, historical corpora usually set up a limited number of longer periods, and assign each text to one of these, so that they can be grouped together for analysis in roughly contemporaneous samples that are large enough to do basic statistics. The YCOE corpus, following the Helsinki Corpus on which it is largely based, divides OE into four periods, the first from the earliest attestations to 850, the second from 850 to 950, the third from 950 to 1050, and the fourth from 1050 to 1150, after which the ME portion of the Helsinki corpus picks up. In what follows, I will collapse together periods one and two because the first contains too little text to be comparable to the others.³² Note furthermore that while the YCOE corpus largely took the OE portion of the Helsinki corpus as a starting point, a number of further texts were added in order to get the total size up to approximately 1.5 million words. These additional texts, unlike those taken over from the Helsinki corpus, were not assigned to one of the four periods. Rather than making a flawed attempt at putting each of these texts into one of the periods myself, I have decided to leave their examples out of consideration for dating purposes, though I do include them in my analysis on all other points. For this reason, the numbers in Table 7 do not add up to the same totals as the numbers in the other tables.³³

With this background, consider now the frequency of preverbal *ge-* for the OE date ranges given in Table 7. The first thing to notice is that the frequencies in the first two periods are remarkably close, indeed essentially identical. In fact, this is the one place where, even with such large numbers, what little difference there is comes out as not being statistically

³²Because I am departing from the periodization in the YCOE, I will indicate the date ranges rather than period numbers, which would only be confusing.

³³Specifically, of the 217,326 total clauses from the YCOE corpus I am considering here, 38,824 (or 17.9%) are left out of the discussion of date, due to having no period assigned in the YCOE.

Dates	ge-	no	% ge-
pre-950	15050	43528	25.7
950–1050	19694	57902	25.4
1050–1150	74	422	14.9

Table 7: Date of the text

significant ($\chi^2 = 1.496$, $p = .2213$). This shows us quite clearly that *ge-* was shockingly stable, neither increasing nor decreasing for most of the OE period. The second thing is that there is a clear and sudden drop in frequency between the second and third periods. At first glance, this suggests that perhaps at this point the decline of *ge-* had begun that would continue through the ME period. However, and this is the third crucial point, we must be very careful about how seriously we take this datum, for two reasons. First, note that the absolute number of clauses we’re looking at here, while not terribly small in absolute terms, is *extremely* small in comparison to the first two periods — two orders of magnitude smaller. This is because of the very limited amount of English text that survives from the relevant period, due to the collapse in the use of written English following the Norman conquest. The number of examples here is not so small that valid statistical reasoning is impossible, but it is small enough that we do have to be concerned about the representativity of the sample.

The second reason to be skeptical is also related to the Norman conquest and the collapse of the Old English scribal tradition. By the early tenth century, a quite consistent, standardized form of West Saxon OE had established itself as the written form used all around the country. Like most standardized languages, it was rather conservative, and by the time of the conquest clearly no longer reflected the contemporary spoken language in many respects. When the scribal tradition was broken by the Norman conquest, the propagation of this standard ceased or was at least severely weakened. Thus, to the extent that people wrote anything in English at this time, they were far more heavily influenced by their own speech than by the inherited standard. This means that there is a sharp break in a wide range of properties of the language we find in the texts of late OE and early ME compared to what came before. But this clearly does **not** imply that there was a series of catastrophic changes in the living language at the time. Rather, there was a catastrophic change in writing practices, such that the written language suddenly caught up with perhaps two centuries’ worth of more gradual changes in the spoken language. In other words, to the extent that the difference between the second and third periods in the table above reflects a real change in the language, it

was probably more gradual than it appears, spread out over the previous century or centuries. What we can conclude with a reasonable amount of certainty is that *ge-* was stable in earlier OE, and was nowhere near disappearing even in very late OE, but that its decline had begun by then.

2.7 The identity of the main verb

The area where the most interesting results are to be found is in the lexical identity of the main verb, i.e. the item on which the prefix *ge-* either does or does not appear. Before we get to the data, a quick word on the specific corpus work it took to get at it is in order. Unfortunately, identifying lexical verbs with searches of the *YCOE* corpus (and for the *PPME2* we'll discuss below) is not nearly as simple as searching for most of the other factors being discussed here. This is because there are vastly more lexical verbs in OE (or of course any language) than there are distinct verb forms, polarity categories, chronological periods etc., and more to the point, because the *YCOE* is not lemmatized. Beyond functional items and a few other extremely common elements — for the verbs this amounts to 'be', 'have', 'do' and the pre-modals — the lexical identity of a word form has not been determined by the corpus creators and is not tagged. The POS-tag on a verb will indicate that it is a verb and provide information about its morphological form and grammatical properties, but not whether it is a form of e.g. *speak* or *eat* or *desire*. This means that identifying specific lexical items requires writing queries that can recognize them based on their forms, which is time-consuming and prone to errors. Different inflected forms of the same verb can have fairly distinct stem forms in OE due to ablaut, umlaut and other irregularities, and the orthography is not always consistent.³⁴ There is an effect of diminishing returns as well due to Zipf's Law. That is, the vast majority of lexical verbs will only appear once or a handful of times in the corpus. Thus it is only really worth the effort of doing the work to recognize a relatively small number of very common items.

That is precisely what I did, writing my queries to recognize 31 lexical verbs based on their forms, in addition to the ones that are specifically tagged in the corpus. This successfully identified 54,380 verb forms with specific verbal lexemes as indicated in the tables below. There were an additional 74,395 verb forms that were not recognized, and these are listed

³⁴In ME the problems actually get worse, as the texts we have reflect a far greater degree of dialectal diversity, and with nothing analogous to the West Saxon standard of OE, the array of distinct orthographical traditions is daunting and impractical for the modern researcher.

below as ‘unclassified’. Table 8 is a complete list of all of the verbs, plus the unclassified category, sorted in ascending order of the percentage of *ge-* vs. no prefix, to give an overview of the situation.³⁵ At the bottom of the table I include the ‘general total’ for comparison, i.e. the overall frequency of *ge-* across all verbs, which is at 25.4%.

Verb	Gloss	<i>ge-</i>	no	% <i>ge-</i>
(pre-)modals		0	2575	0.0
<i>bēon/wesan</i>	‘be’	1	30127	0.0
<i>habban</i>	‘have’	13	5053	0.3
<i>cuman</i>	‘come’	29	4687	0.6
<i>sendan</i>	‘send’	15	947	1.6
<i>drincan</i>	‘drink’	17	779	2.1
<i>etan</i>	‘eat’	26	538	4.6
<i>fēran</i>	‘go’	64	1282	4.8
<i>bēodan</i>	‘offer, proclaim, command’	58	1001	5.5
<i>cwedan</i>	‘say’	553	9145	5.7
<i>gān</i>	‘go’	128	1927	6.2
<i>secgan</i>	‘say’	288	3783	7.1
<i>sprecan</i>	‘speak’	90	1134	7.4
<i>andwyrðan</i>	‘answer’	37	457	7.5
<i>sellan</i>	‘give’	362	2182	14.2
<i>wunian</i>	‘dwell’	202	1093	15.6
<i>writan</i>	‘write’	30	158	16.0
<i>sittan</i>	‘sit’	131	649	16.8
<i>sēoþan</i>	‘boil’	3	14	17.6
<i>fōn</i>	‘grasp’	159	728	17.9
<i>hātan</i>	‘call/order’	560	2309	19.5
<i>dōn</i>	‘do’	933	3681	20.2
<i>slēan</i>	‘smite’	87	325	21.1
<i>faran</i>	‘go’	241	772	23.8
<i>acsian</i>	‘ask’	156	486	24.3
<i>nemnan</i>	‘name’	217	601	26.5
<i>þencan</i>	‘think’	328	777	29.7
<i>wyrðan</i>	‘work, make’	523	1227	29.9
unclassified		30831	43185	41.7

Table 8: Identity of the main verb

³⁵ Again, in computing this percentage, I divided the number of examples with *ge-* by the sum of those with *ge-* and those with no prefix. Examples of these verbs with other prefixes are left out of consideration.

Verb	Gloss	ge-	no	% ge-
<i>tēon</i>	‘pull’	89	119	42.8
<i>weorþan</i>	‘become’	1001	979	50.6
<i>niman</i>	‘take’	1434	1265	53.1
<i>hālgian</i>	‘hallow’	392	108	78.4
<i>hælan</i>	‘heal’	626	110	85.1
<i>sēon</i>	‘see’	2712	159	94.5
general total		42336	124362	25.4

Table 8: Identity of the main verb (continued)

The first, very important message to take away here, is that the variation is massive. It goes from verbs that are literally never prefixed with *ge-* to one that bears it a full 94.5% of the time and fills out the space in between fairly evenly.

Now let’s zoom in a bit to get a better idea of what’s going on in detail, by splitting up that full range of variation into a few smaller chunks. It’s useful to keep in mind throughout that *ge-* appears on the main verb of roughly one quarter of all clauses, thus at the bottom of each of the tables to follow I will always include the ‘general total’ from above for comparison. First, at the very bottom of the range, we’ll take what we can think of as the ‘auxiliary’ verbs. Note that what we’re looking at here are *not* the actual auxiliary uses of these verbs (where *ge-* also never appears), but rather their main verb uses, since in general here we are investigating whether the main verb of a clause bears *ge-*. The frequency of *ge-* with

Verb	Gloss	ge-	no	% ge-
(pre-)modals		0	2575	0.0
<i>bēon/wesan</i>	‘be’	1	30127	0.0
<i>habban</i>	‘have’	13	5053	0.3
general total		42336	124362	25.4

Table 9: Main verb uses of ‘auxiliary’ verbs

these verbs given in Table 9 is essentially zero, and given the very high frequency of their appearance in the corpus, we can be quite confident in the accuracy of this result. Now, given the background assumption from previous work that the *ge-* prefix has something to do with perfectivity or resultativity, such extremely low frequencies are not really surprising. All of these verbs are statives, and so are not expected to appear in perfective

or resultative uses. This is a case where the precise details of what is behind the distribution of *ge-* are perhaps not so crucial, since we expect it to be incompatible with statives under most serious proposals. As we move further into the other verbs on the list, different specific theories will make different predictions, and the ways that specific verb classes behave will help us to choose among them. When considering each group I will continue to initially speak in terms of what is surprising or expected based on a general notion of perfectivity or resultativity, and then make use of the surprises to help lead us toward a specific proposal.

Turning now to the lexical verbs, we consider first in Table 10 the group with markedly low frequency of *ge-*.³⁶ The relative rarity of *ge-* with some

Verb	Gloss	<i>ge-</i>	no	% <i>ge-</i>
<i>cuman</i>	‘come’	29	4687	0.6
<i>sendan</i>	‘send’	15	947	1.6
<i>drincan</i>	‘drink’	17	779	2.1
<i>etan</i>	‘eat’	26	538	4.6
<i>fēran</i>	‘go’	64	1282	4.8
<i>bēodan</i>	‘offer, proclaim, command’	58	1001	5.5
<i>cwedan</i>	‘say’	553	9145	5.7
<i>gān</i>	‘go’	128	1927	6.2
<i>secgan</i>	‘say’	288	3783	7.1
<i>sprecan</i>	‘speak’	90	1134	7.4
<i>andwyrðan</i>	‘answer’	37	457	7.5
general total		42336	124362	25.4

Table 10: Lexical verbs with low frequency of *ge-*

of these verbs is again fairly easy to understand. The group of speech verbs (*bēodan*, *cwedan*, *secgan*, *sprecan* and *andwyrðan*) are arguably all fundamentally activities in Aktionsart terms, meaning that they aren’t telic and thus don’t normally have resultative uses. The two ‘go’ verbs are plausibly also activities, though it will depend here quite a bit on the details of individual contexts. Motion verbs are frequently activities in their

³⁶The cut-off between this group and the next is of course arbitrary. Unlike with the previous group of verbs, there is no clear grammatically defined division here. I have chosen to draw the line between *andwyrðan* and *sellan* as there is a marked jump in frequency of *ge-* between them, from 7.5% to 14.2%. The next cut-off point, between *wyrðan* and *tēon*, also corresponds to a jump in frequency, from 29.9% to 42.8%, and conveniently also separates out the lexical verbs that combine with *ge-* at a higher frequency than the mass of unclassified ones at 41.7%.

basic uses, but — in distinction to a lot of speech verbs — relatively flexible in Aktionsart terms, being easily converted to Accomplishments e.g. by the addition of appropriate PPs indicating a goal. It is thus actually a bit surprising that these ‘go’ verbs have, if anything, a *lower* frequency of *ge-* than the speech verbs, which are far less likely to be made telic by the addition of a PP. We will return to this in Section 3.3.

Initially also unexpected is the behavior of *sendan*, *drincan* and *etan*, which are at the lower end even of this group of verbs with infrequent *ge-*. We would expect these, especially the latter two, to be telic in most cases, and thus if *ge-* marks perfectives or resultatives, it seems that it should be common here. I will come back to *drincan* and *etan* in Section 3.3, where we will see that their behavior can actually provide some support for a particular analysis of the function of *ge-*. With *sendan*, the story is different and a bit simpler. Like with the motion verbs, we expect this to regularly be telic, with an explicit goal provided by an indirect object or PP (*Leela sent Amrit the book/Leela sent the book to Amrit*), hence to take prefix *ge-* if that marks perfectives or resultatives. However, while *ge-* is the most common verbal prefix of its type in OE, and the one whose distribution and meaning present the most challenges, it is really just one member of a larger morphosyntactic system, as alluded to briefly above. It turns out that with *sendan*, other prefixes – specifically *ā-* and *on-* — are overwhelmingly the ones used in telic contexts, not *ge-*. Table 11 provides the numbers for *sendan* with various prefixes vs. with no prefix. We see then that while *sendan*

prefix	freq
none	947
<i>ā-</i>	386
<i>on-</i>	96
<i>ge-</i>	15
<i>tō-</i>	7
<i>for-</i>	6
<i>of-/be-/in-/ūt-</i>	9
total pref.	519
% pref.	35.4

Table 11: Frequency of various prefixes with *sendan*

may have a markedly low frequency with *ge-* in particular, it has a rather normal frequency of prefixation overall for a frequently (but not obligatorily) telic verb.³⁷

³⁷It should be noted here that while I am not systematically considering the other pre-

The really big surprise among the verbs with a low frequency of *ge*-prefixation is with *cuman*. Verbs meaning ‘come’ are typically highly telic — unlike verbs meaning ‘go’, they include an inherent telos.³⁸ Indeed, ‘come’ is typically a telic verb par excellence, and so we expect OE *cuman* to be used primarily in perfective and resultative contexts. Under at least a naïve reading of essentially all accounts of the distribution of *ge*-, we would thus predict a very high frequency with *cuman*. And yet, this particular verb bears the prefix less than one percent of the time, a frequency otherwise found only with statives. In (14) we have a straightforward example, which is clearly telic, perfective, resultative and anything else you might expect to be associated with *ge*-, including being a periphrastic perfect built on the PPP. Nonetheless, the form of *cuman* we find is unprefixated:

- (14) Martha þā gehȳrde þæt sē Hǣlend wæs **cumen**
 Martha then heard that the savior was come
 ‘Then Martha heard that the savior had arrived.’
 (coaelhom,+AHom_6:49.889)

We will return to the status of *cuman*, and how it might be accounted for, in Section 3.3.

Now let’s consider Table 12, which contains the verbs with markedly high frequency of *ge*-, i.e. those which take *ge*- more often than the average of the unclassified lexical verbs. Here again we have both the expected and the unexpected from the perspective of previous attempts to understand the prefix. These are primarily highly telic Achievement or Accomplishment verbs, like *weorþan*, *niman*, *hālgian* and *hǣlan*, which would be expected under all approaches to appear frequently with *ge*-. The surprising case here is *sēon*, which might have been expected to be a stative or an activity at least a significant portion of the time, but in fact overwhelmingly takes *ge*-. As discussed by Karp (1989), what is going on here is that unprefixated *sēon* in OE is best translated as something like ‘look at’, whereas the prefixed form “indicates the moment of perception, of “making one’s

fixes in my examination of the corpus results here, my searches *did* identify them, precisely so that I could exclude them from the count of forms with no prefix at all. I was thus able to check the other lexical verbs identified here with a markedly low frequency of *ge*- to see if any show similar effects to *sendan*. The only one that seems to fit into this category is *fēran* ‘go’, which has an extremely high frequency of the prefix *forþ*: 786 such examples, compared to only 64 with *ge*- and 1282 without a prefix. This appears to be a very specific lexical effect, as *forþfēran* has the idiosyncratic meaning ‘die’ in addition to ‘go forth’ or ‘depart’.

³⁸By default this telos is the location of the speaker at the reference time, but it can be shifted to other salient locations (e.g. the location of the hearer) depending on the context.

Verb	Gloss	ge-	no	% ge-
<i>tēon</i>	‘draw, pull’	89	119	42.8
<i>weorþan</i>	‘become’	1001	979	50.6
<i>niman</i>	‘take’	1434	1265	53.1
<i>hālgian</i>	‘hallow’	392	108	78.4
<i>hælan</i>	‘heal’	626	110	85.1
<i>sēon</i>	‘see’	2712	159	94.5
general total		42336	124362	25.4

Table 12: Lexical verbs with high frequency of *ge-*

own through the immediate act of seeing” [p. 76]. The former is an atelic activity, the latter a highly telic achievement, thus the distribution of *ge-* actually fits quite well with naïve expectations.³⁹

2.8 Interactions with auxiliaries

A final type of factor to consider is the presence of different auxiliary verbs in the clause in addition to the main verb. We can naïvely expect effects on the distribution of *ge-* here because at least some of the relevant periphrastic constructions are used to express aspectual distinctions. Additionally, the presence of particular auxiliaries is extremely easy to search for in the corpora, unlike most other reflections of things related to aspect, so it’s an ideal factor for an investigation focused on automated searches of large amounts of data. Let’s begin then with a comparison of all of the auxiliaries, as well as the possibility of no auxiliary, shown in 13.⁴⁰ Clauses with a (pre-)modal auxiliary have a somewhat higher frequency of *ge-* than the general average, but the difference is not especially large.⁴¹ Auxiliaries

³⁹A different OE verb, *lōcian* (the ancestor of PDE *look*), seems to have had a very similar meaning to the unprefixed *sēon*. It was not common enough in my YCOE samples to make it into the list of verbs that I wrote queries to recognize, so I can’t report on it specifically here. However, Karp (1989) reports that it appears 27 times in *Beowulf*, never with *ge-*, whereas unprefixed *sēon* appears 12 times, and *gesēon* appears 37 times. In other words, unprefixed *sēon* may have been so much less frequent than prefixed *gesēon* because there was a distinct verb with a very similar meaning in competition with the former but not the latter.

⁴⁰The various categories here do not add up to the general total because a number of examples have been set aside where there is more than one auxiliary or the situation is otherwise complicated in a way that makes it difficult to compare to the three main categories here.

⁴¹To be absolutely clear, what is being discussed here is whether, in a clause containing a particular auxiliary, the main verb is prefixed with *ge-*. We are not talking about in-

Aux.	ge-	no	% ge-
none	27834	113607	19.7
(pre-)modal	3374	7442	31.2
BE	9754	2504	79.6
HAVE	969	43	95.8
general total	42336	124362	25.4

Table 13: Presence of auxiliaries in the clause

BE and HAVE, on the other hand, show a very strong favoring effect on *ge-*. Clauses with no auxiliary have a somewhat lower than average frequency with *ge-*, but this is just a necessary consequence of the fact that examples with any type of auxiliary have higher than average frequency.

Now, at first glance it looks like the preference for *ge-* is stronger with HAVE than with BE. However, there's a good bit more going on here that needs to be unpacked. First of all, while auxiliary HAVE is essentially only found in the (ancestor of the) periphrastic perfect, BE is used there as well, but also in the passive and the OE ancestor of the progressive. Aspectually speaking, the passive and especially the progressive are entirely different from the perfect, and we do not expect them to behave at all the same with respect to *ge-*. If we restrict our attention to just perfect clauses, we get the numbers in Table 14.⁴² Now we see that the frequency of *ge-* is

	ge-	no	% ge-
BE	868	96	90
HAVE	125	4	97

Table 14: Perfects, according to auxiliary

much closer to being the same with BE and HAVE, and furthermore that

stances where an auxiliary itself is prefixed with *ge-*. As noted in Section 2.7, this simply does not seem to occur.

⁴²The numbers are much smaller because I have in fact restricted attention to intransitive examples for methodological reasons. Briefly, in the YCOE corpus, PPPs all have the same tag VBN, regardless of whether they appear in a passive, a perfect or some other construction. This means that perfects with auxiliary BE and passives are formally identical and not distinguishable by corpus searches. They must rather be identified on a case by case basis by considering the transitivity of the lexical verb and the semantics of the particular example. Fortunately, this time-consuming hand-coding has already been done, in the research leading up to [McFadden and Alexiadou \(2010\)](#), and I have used that as the basis for the numbers reported here. However, since that work was concerned with the alternation between HAVE and BE in the perfect, and said alternation is restricted to intransitives (transitives always using HAVE), the coding was only done for intransitive examples, yielding the restricted sample reported on here.

it is approaching being categorical. Still, there appears to be a somewhat stronger preference for the prefix with HAVE than with BE.

It turns out, however, that this difference is spurious, and comes entirely from interaction with lexical effects. 86 of the 96 examples of perfects with BE where the PPP lacks *ge-* are with *cuman*. Recall that — for reasons that we haven’t figured out yet — *cuman* staunchly resists prefixation with *ge-*. Crucially, OE *cuman* **only** appears with BE in the perfect in OE (McFadden and Alexiadou, 2010), so the examples with that one verb are artificially suppressing the overall frequency of *ge-* with auxiliary BE. If we remove the examples with *cuman* from consideration, we get the numbers in Table 15. The difference between HAVE and BE is now essentially

	ge-	no	% ge-
BE	861	10	99
HAVE	125	4	97

Table 15: Perfects, according to auxiliary, excluding *cuman*

gone, and we have the effect that once we correct for lexical oddities, *ge-* is more or less categorical with the periphrastic perfect of intransitives in OE.

Consider what this means for our search for an explanation for the function of *ge-* in the language. Since its distribution is so categorical once we carefully distinguish contexts, it is plausible to think that it realizes a single, specific grammatical category, rather than marking a vaguer category that is variably sensitive to multiple factors or being a derivational prefix with the attendant lexical irregularities. Furthermore, this specific category cannot be something that distinguishes the HAVE and BE perfects in OE, since they behave identically. It must rather be a component that all OE perfects have in common, which however is not limited to or diagnostic of the perfect, as it is present at a relatively high frequency in non-perfect clauses as well.

3 An analysis of the OE patterns and some explanation

3.1 An excursus on the semantics of early English ‘perfects’

I would like to suggest that the facts just discussed from the perfect provide a key to unlocking the mysterious role of *ge-* in OE. In particular, they

lead quite directly to the proposal that the prefix has something to do with resultativity rather than perfectivity. In order to motivate this we need to first consider some background on the semantics of the ‘perfect’ in the various stages of early English. Part of what makes the ‘perfect’ difficult to get a handle on is that there seem to be multiple readings for it, which are distinguishable in terms of their entailments, yet can be expressed by the same morphosyntactic form, at least in many languages (see [Iatridou et al., 2003](#), and the other contributions in [Alexiadou et al., 2003](#) for useful discussion). Indeed, what we find in historical stages of English was crucially different from PDE, with a series of changes occurring especially in ME and in subsequent periods. This is why I use scare quotes around ‘perfect’, to make clear that we are not dealing with a single unified phenomenon across time.

For present purposes we need to be able to distinguish between the readings referred to in the literature as the **perfect of result** and the **experiential perfect**, both of which can be expressed by the periphrasis with auxiliary HAVE in PDE. The perfect of result entails that the target state of the eventuality described by the main predicate holds at the reference time. In (15), e.g. the continuation makes it clear that we’re not just talking about what Beorhtric has done, but what state he is currently in as a result of what he has done. I.e. he is in York at the reference time (which happens to also be the speech time because this is a present perfect) as a result of going there.

(15) Beorhtric has gone to York, and he won’t be back until tomorrow.

(16) Beorhtric has gone to York five times already this semester.

The experiential perfect, on the other hand, entails that the eventuality described by the main predicate is anterior to the reference time. There are not necessarily any implications about whether or not particular consequences of that eventuality continue to hold, aside from the somewhat trivial fact that what has happened cannot un-happen, i.e. the subject will always have the experience of having participated in the eventuality, even if a potential target state of that eventuality no longer holds.⁴³ Example (16) must be interpreted as an experiential perfect because of the repetition involved. It is not possible for the target state of at least the first four instances of Beorhtric going to York to still hold, since he must have left in the meantime in order to be able to go there again. That is, there is clearly no implication that Beorhtric is in York five times (or even that he is in

⁴³In [Parsons \(1990\)](#)’s terms, the resultant state holds, but there is no implication about a target state.

York at all) at the reference time, which would be incoherent, but rather that he is in the (resultant) state of having experienced going to York five times in the current semester.

Now, it is well established that the ‘perfect’ constructions in OE, built with BE and HAVE in addition to the PPP, crucially differ from their PDE descendants in that they were essentially restricted to the perfect of result (see [McFadden and Alexiadou, 2010](#), and citations there). The experiential reading was not yet available. In other words, while you could use a periphrastic ‘perfect’ to express something like (15) in OE, you could not use it to express (16). A simple past form would have been used instead. This means that every time we see a perfect in an OE text, we can conclude that it is a resultative, i.e. there is a target state that is asserted to hold at the reference time. This target state is then something that the perfects with HAVE and BE in OE all have in common, and thus a candidate for what *ge-* contributes. Indeed, we can go further, since it is certainly not the case that target states are only found in perfects. Rather, target states can be found in clauses with all kinds of tense and Viewpoint Aspect, which is at least broadly in line with the distribution we observe for *ge-*.

3.2 The proposal and its implementation

I would thus like to propose and explore the hypothesis that OE *ge-* marks a particular type of resultativity, being associated with the presence of a target state. This of course has clear connections to some of the earlier proposals mentioned above, including [Ramchand and Svenonius \(2002\)](#); [McIntyre \(2003\)](#); [van Kemenade and Los \(2003\)](#); [Los et al. \(2012\)](#). Above all, it connects to [Karp \(1989\)](#), differing only in being slightly more explicit about the type of resultative, i.e. involving a target state as opposed to a resultant state.⁴⁴ It is clearly distinct, on the other hand, from proposals that connect *ge-* to Viewpoint Aspect, in particular [Streitberg \(1891\)](#)’s claim that it marks perfectivity. The relationship to [Lloyd \(1979\)](#)’s proposal that its Gothic cognate marks ‘complexive’ aspect is a bit more complicated, since his definition of complexive seems to involve what I have defined as Viewpoint Aspect with a touch of Aktionsart, as I will discuss below.

The idea being pursued here thus associates *ge-* with Aktionsart, rather than with Viewpoint Aspect. Note, however, that I am **not** claiming that *ge-* marks a specific (Vendlerian) Aktionsart, but rather one of the building blocks that goes into at least two different ones. Modern work on

⁴⁴My reading of [Karp \(1989\)](#) is that she meant resultativity in this sense as well, but she did not lay it out explicitly in these (or clearly equivalent) terms.

Aktionsart, even if it adopts the four (or five) main categories going back to Vendler (1957), typically decomposes them in terms of more basic elements (see Dowty, 1979, among many others), and the target state is one of these. It is a defining ingredient of the telic Aktionsarten, i.e. of Achievements and Accomplishments, distinguishing them from Activities, States and Semelfactives).

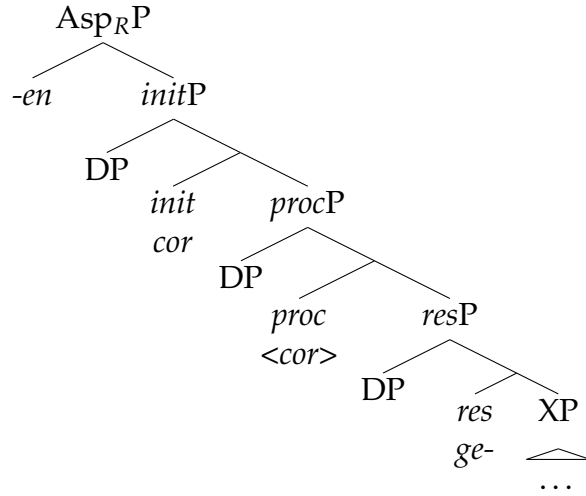
What I would like to argue for now is that the OE data discussed so far actually allow us to be even more specific about what is meant by saying that *ge-* is associated with resultativity and target states. I will implement my proposal in terms of Ramchand (2008)'s verbal decomposition, which identifies the basic building blocks of the temporal and eventive structure of predicates with dedicated structures in the syntax. She posits three main heads in what she dubs the 'first phase'. The middle head, called *proc(ess)* encodes a (durative) process. The higher head, called *init(iation)* encodes a state which causes the process, i.e. the state of affairs that sets the process in motion, like the intention of an agent. The lower head, called *res(ult)* encodes a state that is caused to hold by the process, i.e. the target state of the complex eventuality. The Aktionsart of an eventuality is determined in large part based on which of these heads are present. For example, a typical Activity will include *proc* but not *res*, and may or may not include *init*, depending on whether it is externally caused. An agentive Achievement, on the other hand, will include all three heads, with *init* encoding the agentive causation, *proc* the process portion, leading up to the target state represented by *res*.⁴⁵

Given this background, I would like to propose that *ge-* is the default realization in OE of Ramchand's *res* head. The *-en* or *-d* suffix in PPPs is higher up, in an Asp head outside of *initP* (Kratzer, 2000; Embick, 2004; McFadden and Alexiadou, 2010). Concretely, for a form like the PPP *gecoren* 'chosen' in a perfect clause like example (11) above, repeated here as (17), we have the structure in (18):

- (17) ac hēo hāfde **gecoren** Crist hyre tō brȳdguman
 but she had chosen Christ her to bridegroom
 '...but she had chosen Christ as her bridegroom.'
 (coalive,+ALS_[Eugenia]:349.401)

⁴⁵Ramchand (2008) argues that traditional Accomplishments differ from Achievements, somewhat unexpectedly from a naïve perspective, in lacking the *res* head. Their boundedness or telicity is contributed instead by incremental theme or PATH complements, a point we will return to below as it turns out to be crucial for our analysis of some of the initially surprising facts about OE *ge-*.

(18)



In addition to the basic semantics (about which more below), this structural proposal accounts for certain simple morphological facts, e.g. that *ge-* is a prefix while *-en* is a suffix. Given the Mirror Principle (Baker, 1985), if the four heads in the structure above combine in a single word, the expected unmarked order will be *res-proc-init-Asp_R*, which is exactly what we get.⁴⁶ Note that it is cross-linguistically common for resultative elements to show up as verbal prefixes, and so this seems like a quite reasonable result.

This covers the most basic facts of the distribution of *ge-*, including the alternation with specific lexical verbs between forms with and without the prefix. The relevant verbal roots on their own are able to lexicalize the *proc* head and potentially also the *init* head, i.e. at base they describe Activities. In order to derive a telic Achievement, the *res* head is necessary too, which these verbs are not specified to realize. The default spell-out of *res*, *ge-* is thus inserted instead, deriving the correlation between the appearance of *ge-* and resultativity.

3.3 Covering the data

Let us consider now how this proposal can accommodate some of the surprising data that we turned up for specific lexical verbs. First, how do we deal with the extreme dispreference for *ge-* with *cuman*? Note that the pattern here is out of line with expectations no matter what approach we take to the basic semantics of *ge-*. As noted above, ‘come’ is clearly a telic

⁴⁶The *init* and *proc* heads are jointly realized by the verb root, hinted at here by having *cor* start in *proc* and move up to *init*. See Ramchand (2008) for an approach to this kind of morphological realization in terms of Spanning.

predicate, with a strong resultative component, and is certainly expected to appear frequently in perfective uses. The idea being pursued here, that *ge-* is associated with target states realized as Ramchand's *res* head, fares no better on its own, since the semantics of 'come' clearly *does* include a target state. What this means is that we need a theory that can treat *cuman* as some kind of lexical exception, and ideally also make sense of why this of all verbs should be exceptional in this way. Karp (1989) proposes essentially this, analyzing *cuman* and a handful of other verbs as "resultative-simplexes", which "never appear with the *ge-* prefix because their meaning is already resultative" [p. 18]. While the proposal being made here cannot claim to be uniquely suited in this sense — any reasonable theory will have a mechanism for dealing with lexical exceptions of this kind — the solution it offers is at least more than adequate.

The relevant bit of background is that Ramchand's system allows for individual verbal items to simultaneously realize multiple head positions, subject to lexical restrictions. That is, some lexical verbs can be specified to realize *init* + *proc*, others *proc* + *res* or *init* + *proc* + *res*, and still others just *proc*. Indeed, in a language like PDE, a large number of lexical verbs can realize the entire spine of the first phase, including two or three of the relevant functional heads, without any help from additional morphology. What I am proposing for OE in contrast is that, as a language-specific property, its lexical verbs are typically not specified to realize the *res* head, which thus must be spelled out separately from the verb root in structures in which it appears, and furthermore that precisely this is what the prefix *ge-* does. Now, to deal with a verb like *cuman*, we can simply posit that it is exceptional in that the verbal root itself realizes the *res* head in addition to *init* and *proc*. I.e. its lexical entry contains the specification [*init*, *proc*, *res*]. Under the principles Ramchand adopts for how lexical items compete to realize particular bits of structure, an element like *cuman* will span across all three heads, winning out over and thus blocking *ge-*. This of course works technically and covers the empirical ground we need it to cover, but it can be reasonably argued that it goes beyond this minimum to be a fairly natural account of the situation and to actually help make sense of why this particular verb should behave this way.

First of all, in Ramchand's system, what determines what spells out the different heads in the first phase is the lexical information specified for particular verbal items, interacting with general principles for resolving competition. If *ge-* is competing with lexical material to spell out a particular head in that system, it is entirely expected that there will be some lexical exceptions where it loses the competition, as we find with *cuman*. Second of all, the specification of what heads in the first phase a lexical

item can realize can be thought of as essentially a grammaticalization of its typical semantic behavior. Thus if there are going to be lexical items that are exceptionally specified to be able to realize the *res* head, we would expect them to be precisely those lexical items that (more or less) exclusively appear with such semantics. In other words, we predict that a verb like *cuman*, which is basically always used in resultative contexts, will be able to supercede *ge-*, i.e. to be inherently resultative, while a verb like, say, *faran* ‘go’, which is sometimes used in resultative contexts and sometimes in non-resultative ones, will not do so.

What then about the markedly low frequency of *ge-* that we noted with *etan* and *drincan*? These are verbs that may not always involve a clear target state, but certainly will much of the time, when a clearly defined substance ends up being consumed. Consider an example where both of them happen to appear, and where there are clear target states defined for the definite objects they take:

- (19) þā sacerdas þā ... æton þone mete ... and þæt wīn eal
the priests then ... ate the food ... and the wine all
druncon
drank
‘The priests then ate the food and drank all the wine.’
(coaelhom,+AHom_22:406.3519-21)

Given everything we have said so far, we certainly expect the result states of the food and all the wine being consumed to be reflected by the presence of *ge-* on the two verbs.

As it turns out, however, [Ramchand \(2008\)](#)’s theory actually predicts this pattern. She makes a distinction between resultative meanings that come from the structural specification of an actual target state, and those that arise from the presence of a bounded PATH or incremental theme argument. While the former involve an explicit *res* head in the structure, the latter do not, with the resultative meaning instead being an entailment of how the rhematic material restricts the interpretation of *proc*. That is, they do not actually involve a *res* head. Note then that clauses built around *etan* and *drincan*, as consumption verbs, will primarily be found in structures of this latter type. They realize *init* and *proc*, and combine with incremental themes like *þone mete* and *þæt wīn* above, which measure out and bound the process, providing an implication of telicity. That is, it is possible at any time to gauge the priests’ progress in completing the eventuality described by examining how much of the food and wine are gone. However, no *res* head will be involved in these structures, and thus there is no place for *ge-* to be inserted. Hence we can account for why *ge-* is generally not

found with these verbs, even though they have a ‘resultative’ interpretation in a pre-theoretical sense. This is a clear advantage over competing proposals in terms of telicity or a less precisely defined resultativity. [Karp \(1989\)](#) e.g. classes these together with *cuman* as simplex-resultatives, but it seems to me that this misses an important contrast: verbs like *etan* and *drincan* are not expected to *always* be telic with a clear target state, and in fact in the absence of a definite DP object representing an incremental theme, they default to describing atelic Activities. Thus the approach we adopted for *cuman* does not seem appropriate.

This same account can be extended to motion verbs like *faran*, *fēran* and *gān*. Fundamentally, these describe unbounded Activities, again quite distinct from *cuman* with its inherent telos. Eventualities involving such motion verbs are frequently telic, involving a goal or a scalar PATH that behaves analogously to an incremental theme. But these again do not involve the *res* head, instead being supplied by PPs or adverbials in the complement of *proc*. We thus again have a way of producing structures that are telic or resultative in a clear descriptive sense, yet which do not involve the *res* head, and thus are not associated with *ge-* prefixation.

I propose that this class of verbs provides an argument in favor of the kind of resultativity as a component of Aktionsart approach being pursued here over Viewpoint Aspect approaches like [Streitberg \(1891\)](#); [Lloyd \(1979\)](#) etc. VPs built around consumption and motion predicates can clearly be telic, and so it is difficult to see why they should be incompatible with a perfective Viewpoint Aspect that depends on completion. [Lloyd \(1979\)](#)’s answer to this, e.g. for why Gothic *qiman* ‘come’ doesn’t appear with *ga-*, is that it is a ‘punctual’, and these are incompatible with the complexive. Punctuals describe only the transition from one state to another, with no implication whatsoever of an Activity or Event leading up to that transition. But for Lloyd, a marked complexive is only motivated in order to describe a complete change that follows from a non-trivial process, because only this level of complexity motivates the Viewpoint-Aspectual omniscience of describing both the boundedness of the eventuality from the outside and its internal structure. If there is a punctual change without any lead-up, the simpler constative is entirely sufficient. This is a coherent position to take, and I am not in a position to challenge its applicability to the Gothic material, but I think that it unnecessarily complicates the picture, at least for OE, by commingling Viewpoint Aspect with Aktionsart. What is more, it runs into trouble in Slavic, as discussed in [1.4](#), where the perfective *can* be used with punctuals, forcing Lloyd to do some fancy footwork in the way that complexive is defined and interacts with the rest of a linguistic system in order to cover Gothic and Slavic

equally.

4 ME developments

An additional point that makes the analysis of *ge-* in OE just presented particularly attractive is that it offers insights into its development in ME. In this section I report on results from searches on the complete PPCME2 (Kroch and Taylor, 1999), which is analogous to the YCOE discussed above: also containing 1.5 million words, tagged for part of speech and grammatical features and parsed for syntactic structure (using a related and extremely similar tagset), but covering ME rather than OE. The corpus work on ME adopted precisely the same methodological strategies as that on OE, described in Section 2.1 above.⁴⁷ Not surprisingly, the use of the prefix shows a steady decline over the course of late OE and ME, as we can see if we extend Table 7 from Section 2.6 above into ME, as shown in Table 16.⁴⁸ Again, the frequency of *ge-* is impressively stable in the first two pe-

Dates	<i>ge-</i>	no-pref	total	<i>ge</i> -%
pre-950	15050	43528	58578	25.7%
950–1050	19694	57902	77596	25.4%
1050–1150	74	422	496	14.9%
1150–1250	2297	30190	32487	7.1%
1250–1350	989	16850	17839	5.5%
1350–1420	1106	58519	59625	1.9%
1420–1500	162	31614	31776	0.5%

Table 16: Frequency of *ge-* in OE and ME

riods of OE, but then begins to drop in the final OE period. While I noted above that we have to be careful about taking the numbers from this period too seriously given the relative paucity of examples, we can see now that they do fit in well with subsequent developments in ME. The overall frequency of *ge-* is significantly lower than in OE from the beginning of the ME period, and by the end of ME, the prefix has essentially been lost, aside from a small number of relics.

⁴⁷The only differences were at the level of details in the queries, targeting points of difference between the two corpora and the underlying languages.

⁴⁸I am here adopting unaltered the four periods distinguished for ME in the PPCME2 corpus, but am again referring to them in terms of year ranges rather than period numbers to keep things parallel with what I have done for OE.

None of the theories on the market for the function of *ge-* in OE offers any real clue about what might underlie this decline. We certainly cannot expect speakers to have had less need of resultativity or of perfectivity or any of the other proposed categories. They certainly won't have stopped using *ge-* because they stopped talking about target states. One possibility, given the proposal made for OE *cuman* above, is that lexical verbs increasingly came to be able to realize the *res* head themselves, obviating the need for a separate realization by *ge-*. At some level this must ultimately be the case, since in Ramchand's system, verb roots that can appear in Achievements in PDE must be analyzed as covering the *res* head. But this just pushes the question one step further down the line: what led (certain) verb roots to expand their realization in this way, at the expense of the inherited prefix *ge-*?

Perhaps the most likely explanation is a relatively mundane morpho-phonological one, with developments akin to what happened in the pre-history of Old Norse. The prefix was unstressed and had relatively little articulatory-phonological substance to begin with, starting as /jε/ in early OE — a glide followed by an unstressed, short vowel. In late OE it would have been reduced to /jə/ by regular sound changes affecting unstressed vowels, and by ME, where the usual spelling has become *i-* or *y-*, it was simply /i/ or perhaps even /ə/. While it probably should not have completely disappeared due solely to sound change, it would have been reduced enough to plausibly be susceptible to morphologically conditioned loss. Note that two of the other prefixes that have survived into the modern language — *be-* and *for-* — had more phonological substance, beginning with an obstruent, although even they have seen their distribution heavily reduced. See [Arista \(2012\)](#) for more discussion.

Even if we can't be sure about why *ge-* disappeared, the way in which it did so yields insights into what it had been doing beforehand. Its decline in frequency is not uniform across environments, but proceeds differently in the two places where *ge-* is most common — perfects and passives, as shown in Table 17. Note that in the first period of ME, the frequency of *ge-*

Dates	Perfect				Passive			
	<i>ge-</i>	no-pref	total	<i>ge-</i> %	<i>ge-</i>	no-pref	total	<i>ge-</i> %
1150–1250	437	424	861	50.75%	967	1222	2189	44.18%
1250–1350	217	265	482	45.02%	352	1096	1448	24.31%
1350–1420	213	1891	2104	10.12%	691	4730	5421	12.75%
1420–1500	10	1247	1257	0.80%	85	3136	3221	2.64%

Table 17: *ge-* in perfects and passives in ME

is comparable in the two environments. But while the drop in the passive is fairly smooth over the next three periods, in the perfect the frequency remains relatively stable into the second, before dropping precipitously in the third ME period.

We can actually make sense of this difference in development if we consider the resultative analysis of *ge-* being proposed here in light of [McFadden and Alexiadou \(2010\)](#)'s findings on the development of the perfect in ME. As noted above, they showed that in OE and early ME, the periphrastic perfect was only used with a perfect-of-result reading, and thus could only be built on resultative predicates. Starting in the period from 1350 to 1420, however, new experiential — crucially non-resultative — uses of the perfect with *have* (but not with *be*) arose. Again, this accounts for why *ge-* was so common in the perfect in the early periods if, as proposed here it was the default realization of the underlying resultative structure. More importantly for current purposes, it also predicts the sudden drop in the frequency of *ge-* in perfects, precisely in the period from 1350 to 1420, due to the influx of the new experiential perfect. The experiential perfect placed no resultativity requirement on the predicates it was built on, thus did not favor *ge-* the way the old resultative perfect had. Indeed, as [Table 17](#) shows, the marked decrease in the percentage of perfects with *ge-* in that period results not from a decrease in instances of *ge-*, but from a sudden increase in the total number of perfects, just as expected. The old perfect-of-result, which favored *ge-*, continued to be used at similar rates as before, but was swamped by the new experiential one, which did not favor *ge-*. This is a striking parallel to [McFadden and Alexiadou \(2010\)](#)'s finding that the purely resultative *be*-perfect was swamped by the resultative-or-experiential *have*-perfect in the same period.

5 Conclusion and outlook

In this paper I have presented data and conclusions from a large-scale corpus study on the use of preverbal *ge-* in Old and Middle English. On the basis of the evidence obtained, I was able to propose an analysis of the prefix which is more explicit about the specific aspectual components involved than previous approaches, and which can cover certain otherwise puzzling factors of its distribution. As noted at the outset, the work here remains incomplete in an important sense: the corpus study has so far only involved automated searches, which means that only those patterns have been investigated that can be unambiguously identified on the basis of the parsed structure or annotation in the corpora, or on the basis of

specific string forms. As a result, a wide array of syntactic and especially interpretive factors have not yet been taken into account.

This was justified by a strategy meant to make it possible to deal with the uncommonly large volume of data associated with the distribution of *ge-* — it is simply too large for it to be feasible to examine all of the examples by hand. This description of the issue already suggests how it can be addressed in future work. A representative sample of a manageable size needs to be extracted from the collection of data made so far, and the examples there examined and coded by hand for factors that could not be automatically searched for. In particular, interpretive and other subtle factors that can be used to diagnose categories of Viewpoint Aspect and Aktionsart can be specifically targeted, in order to test the account put forward here against competitors that tie *ge-* to Viewpoint Aspect or other grammatical or semantic categories.

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